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Home range and habitat use of male ruffed grouse in managed mixed oak and aspen forests

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Abstract

Ruffed grouse (*Bonasa umbellus*) occupy many areas of the eastern and southeastern United States where oak (*Quercus* sp.) is the dominant forest type. However, most research on grouse habitat requirements and populations has taken place in Lake States aspen (*Populus* sp.) forests. We captured and radiotagged male grouse to determine spring and summer home range sizes and habitat use on a 1120 ha experimental forest management area in central Pennsylvania. The dominant forest types were aspen-black cherry and mixed oak. The management area was divided into two sections, one which was managed with a short-rotation clearcutting scheme and one that was left unmanaged. Breeding season and summer home ranges of male ruffed grouse were similar to those observed in the Midwest (5.0–9.4 ha), and did not appear to differ between managed and unmanaged habitats. However, habitats used differed between the managed and unmanaged areas. Male grouse on the managed area used aspen-black cherry and mixed oak sapling stands more than expected during both the breeding season and summer. Pole-sized and sawtimber aspen stands in the unmanaged area were used more than expected during breeding and summer seasons. Male ruffed grouse avoided mixed oak sawtimber stands, which comprised 62% and 47% of the available habitat in the unmanaged area and managed area, respectively. Our results demonstrated that southern mixed oak stands can be managed effectively for ruffed grouse with small (1 ha) clearcuts. © 1998 Elsevier Science B.V.

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1. Introduction

Ruffed grouse are closely associated with early-successional forests (Bump et al., 1947; Gullion et al., 1962), and aspen types are regarded as the most

productive grouse habitats (Gullion, 1977; Thompson and Fritzell, 1989). In aspen forests, small clearcuts that create a diversity of forest size-classes in a small area have been demonstrated to increase ruffed grouse populations (Gullion and Svoboda, 1972; Gullion and Alm, 1983; Kubisiak, 1985). However, much of the southeastern portion of ruffed grouse range is primarily oak forests that lack aspen (Cole and Dimmick, 1991). Ruffed grouse in southern areas tend to use young forest stands similar in structure to aspen

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forests (Hunyadi, 1978; Stoll et al., 1979; Hale et al., 1982; Thompson and Fritzell, 1989; Wiggers et al., 1992) but occur at lower densities than in aspen dominated forests.

In recent decades, several states in the southern portion of ruffed grouse range have attempted to reintroduce or increase existing grouse populations (Gudlin and Dimmick, 1984; Widner et al., 1988; Cole and Dimmick, 1991). However, there is little information regarding management of oak forests for ruffed grouse and the response of ruffed grouse to such management. We studied the spatial aspects of male ruffed grouse behavior in managed and unmanaged forests during 1990 and 1991 in central Pennsylvania.

We studied only male grouse because males advertise their presence during the spring mating season and counts of drumming males are a widely-used population index. The density of breeding males has been shown to be highly correlated with the presence of early-successional stands in aspen forests (Gullion et al., 1962). The objective of the overall habitat management plan was to determine if creation of early successional stands in oak forests would increase ruffed grouse density (Liscinsky, 1980). We thought it was important to determine if breeding males, the focus of the population surveys, used managed oak forests differently than unmanaged forests. Our objectives were to: (1) determine breeding season and summer home range size for male ruffed grouse, and (2) determine the extent that male ruffed grouse used aspen and mixed oak forests managed with small (≤ 1 ha) clearcuts during the breeding season and summer.

2. Study area

The study was conducted on State Game Lands (SGL) 176 (40°50'N, 77°53'W), 8 km northwest of State College, Pennsylvania where the Pennsylvania Game Commission (PGC) initiated an intensive forest management study for ruffed grouse in 1975 (Schmaltz, 1978). Topography consisted of two low ridges surrounding a broad valley (Clarke, 1946); elevation ranged from 360 m in the valley to 450 m on ridgetops. Soils were typically sandy and almost entirely of the Morrison series (Braker, 1981). The

transition between aspen-scrub oak and mixed oak types was generally sharp; mixed oak occurred on the slopes and ridges with aspen-scrub oak dominant in the valley (McDonald, 1993).

The 1120 ha study area was divided into a 576 ha unit that received no management (untreated area) and a 544 ha unit that was to be intensively managed (treated area). The treated area was divided into 136, 4 ha blocks. These blocks were subdivided into 4, 1 ha patches designated A–D (Fig. 1). Each block was classified as being in either aspen-scrub oak or mixed oak forest (Schmaltz, 1978).

The 1 ha patches were scheduled to be clearcut on a rotation according to position (A–D) in the 4 ha block and the cover type of the block. Aspen-scrub oak patches were to be initially clearcut on a 20-year rotation and mixed oak patches on a 40-year rotation (Schmaltz, 1978). For example, in each aspen-scrub oak block 1 patch would be cut every 5 years starting with the A-patch. The mixed oak blocks would have 1 patch cut every 10 years. At the end of the first cutting cycle the rotations will be increased to 40 years in the aspen-scrub oak type and 80 years in the mixed oak type. This harvest schedule would create a mix of forest age classes within each block that corresponded with the longevity of the trees and the habitat requirements of ruffed grouse (Gullion, 1972).

All 136 A-patches in aspen-scrub oak and mixed oak types were cut in winter 1976–1977. Sixty B-patches (aspen-scrub oak only) were cut in winter 1981–1982. From 1985 to 1987, 76 B-patches (mixed oak) and 60 C-patches (aspen-scrub oak) were cut. Thus, between 1975 and 1991, 3 of 4 (75%) patches in each of 60 aspen-scrub oak blocks and 2 of 4 (50%) patches in each of 76 mixed oak blocks had been clearcut.

Pure stands of scrub oak occurred only in the sapling size-class. Scrub oak occurred widely in the understory in other forest types. American hazel (*Corylus americana*) occurred in the understory in many areas, sometimes to the exclusion of other shrub species (McDonald, 1993). Black cherry was a major component of both aspen and oak associations. Pitch pine (*Pinus rigida*) comprised nearly all of the pine on the area. However, some scattered areas of white pine (*P. strobus*) were present near the northern boundary of the treated area.

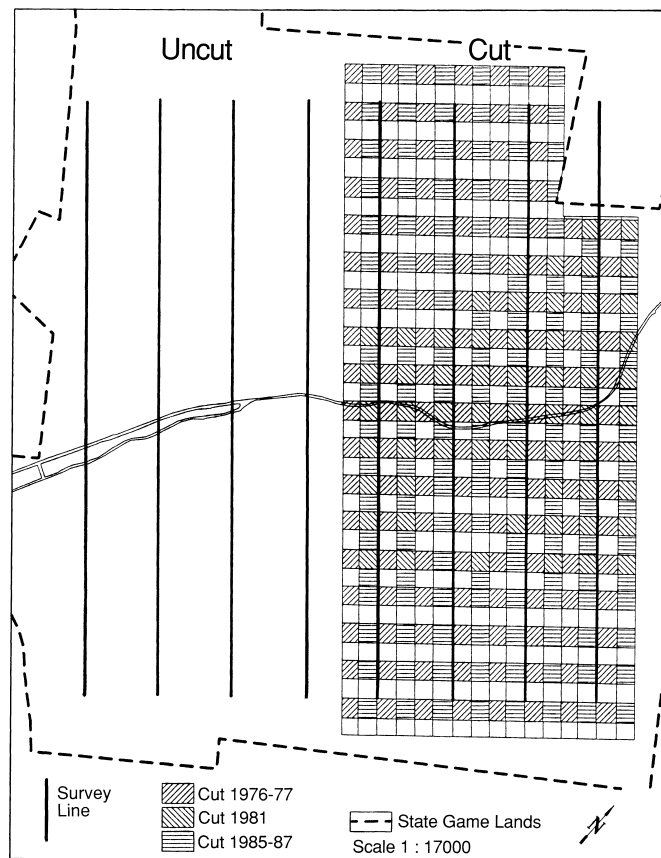


Fig. 1. Layout of the study area and status of the treated (i.e. clearcut) blocks during 1990–91, State Game Lands 176, Pennsylvania.

3. Methods

3.1. Capture and telemetry

We captured male ruffed grouse during spring with mirror traps set on drumming logs (Bowers, 1948). We determined sex and age of captured grouse using procedures described by Davis (1969); Roussel and Ouellet (1975). We weighed grouse with spring scales to the nearest grams. We fitted each grouse with an 11–17 g necklace or poncho style (Small and Rusch, 1985) transmitter in the 164 MHz range. Transmitter packages were $<3\%$ bodyweight of each grouse. We banded each grouse with a numbered aluminum leg band on each leg.

We located each grouse once daily from time of capture until August 31, transmitter failure, or death of the individual, with a portable radio receiver

and a hand-held, 3-element, Yagi antenna. We used the homing technique (Mech, 1983: 60–61) to locate radiotagged grouse. A series of 8 survey transects, 3.2 km long and 0.42 km apart with numbered stations at 0.42 km intervals existed on the study area (McDonald et al., 1994). Thus, radiotagged grouse could be circled on the untreated area and their position fixed by pacing along a perpendicular bearing to the nearest transect and station. On the treated area, the cut and uncut patches, along with the transects enabled us to accurately locate grouse within an area <0.25 ha. Locations were plotted on 1:5000 scale maps and recorded as Universal Transverse Mercator coordinates to the nearest 10 m. We attempted to flush grouse if they did not appear to change location for 2 days or if they moved an unusually great distance (>200 m) between consecutive locations.

Tracking days were divided into 4, 4 h periods: 0501–0900, 0901–1300, 1301–1700, and 1701–2100. We located each grouse approximately an equal number of times during each period each month.

3.2. Home range

Seasons were defined as breeding (capture date – May 31) and summer (June 1–August 31 or loss of contact prior to August 31). We used the McPaal computer program (M. Stuwe, Nat. Zool. Park, Front Royal, VA, unpub. manual) to calculate seasonal and monthly home ranges based on the minimum convex polygon [MCP (Mohr, 1947)] and harmonic mean (Dixon and Chapman, 1980) methods. We calculated 95% harmonic mean home ranges using 10 grid cell divisions (121 intersections). We calculated breeding season home ranges only for grouse that survived ≥ 21 days after capture.

We compared home range estimates within and between the treated and untreated areas using *t*-tests. The significance level for all tests was $P=0.10$. We chose an alpha level of 0.10 to increase the power of the tests (Cohen, 1988). Because a relatively small number of grouse were captured and survived on each area we think alpha=0.10 was a reasonable compromise between Type I and II errors.

3.3. Habitat use

We used color aerial photographs taken in 1990 and 1991 (scale=1:2800) to delineate 12 habitat types on the study area (Table 1). Forest stands were delineated according to species associations and size [sapling, 2.5–12.5 cm diameter at breast height (dbh); pole-timber, 12.6–27.9 cm dbh; sawtimber, ≥ 28.0 cm dbh]. Stands of the same size were combined within species groups regardless of tree density, with the exception of sparse (40–50% canopy cover) sapling stands.

We calculated the amount of each of 11 habitat types to determine habitat availability. We calculated habitat availability separately for the treated and untreated areas. Because the treated and untreated areas were adjacent and grouse could move between them, we included a 200 m buffer zone of each area (treated, untreated) in the availability calculations for the other area.

Table 1

Area^a (ha) in each defined cover type on treated and untreated areas of State Game Lands 176, Pennsylvania, 1990–1991

Cover type	Treated	Untreated
Urban-built up ^b	5.8	15.3
Grassy-open	33.6	33.3
Aspen-black cherry saplings ^c	103.9	6.3
Aspen-black cherry sparse saplings	24.5	9.0
Aspen-black cherry poletimber	47.5	109.0
Aspen-black cherry sawtimber	8.7	19.2
Scrub oak ^d	30.8	41.9
Mixed oak saplings	80.4	9.7
Mixed oak sparse saplings	51.0	6.9
Mixed oak poletimber	13.9	6.1
Mixed oak sawtimber	376.0	479.8
Pine-mixed forest	27.5	58.3
Total	803.6	794.8

^aAreas in each cover type on the treated and untreated area include a 200 m buffer zone.

^bUrban-built up type excluded from habitat use analysis.

^cSapling, 2.5–12.5 cm dbh; poletimber, 12.6–27.9 cm dbh; sawtimber, ≥ 28.0 cm dbh.

^dAll scrub oak forests were sapling size.

We used the X^2 goodness-of-fit test (Neu et al., 1974) to assess grouse use of habitats. We used Bonferroni confidence interval estimation (Neu et al., 1974), with a family-wide error rate of $P=0.05$, to determine preference of habitat types. Because locations were pooled within the treated and untreated areas sample sizes were large enough in each month and season to set alpha=0.05.

4. Results

We captured 15 male grouse from April 28 to May 13, 1990 and 19 from March 29 to May 15, 1991. We estimated home ranges for 5 grouse in 1990 and 14 grouse in 1991 that survived ≥ 21 days after capture. We recorded 1,741 locations (958-treated, 783-untreated) for these 19 grouse. We pooled data from 1990 and 1991 within the treated and untreated areas for home range and habitat use comparisons.

4.1. Home range

Mean MCP and harmonic mean home range estimates did not differ ($P>0.10$) between the treated area and untreated area within any month or season

Table 2

Seasonal minimum convex polygon (MCP) and 95% harmonic mean home range estimates (ha) for male ruffed grouse on treated and untreated areas of State Game Lands 176, Pennsylvania, 1990 and 1991

Season	Treated area					Untreated area				
	<i>n</i>	MCP		Harmonic mean		<i>n</i>	MCP		Harmonic mean	
		$\bar{x}$	SE	$\bar{x}$	SE		$\bar{x}$	SE	$\bar{x}$	SE
Breeding ^a	11	5.0	1.2	6.6	0.9	8	7.2	1.6	9.4	2.3
June	11	5.5	1.7	5.6	1.4	8	6.1	1.0	7.7	1.4
July	9	5.6	1.1	7.3	1.6	8	5.8	1.5	7.4	1.9
August	6	4.4	1.3	5.3	1.2	7	6.0	2.0	8.6	3.3
Summer ^b	11	11.3	2.8	11.6	2.4	8	12.2	2.1	13.6	2.3

^aCapture – May 31.

^bJune 1–August 31 or earlier loss of contact.

(Table 2). However, the small samples of radiotagged male grouse resulted in low power tests of the hypothesis of no difference between areas (MCP: power=0.06–0.28; harmonic mean: power=0.06–0.32). Breeding season home range estimates ranged from 1.7 to 15.6 ha on the treated area and from 2.8 to 22.1 ha on the untreated area. Summer home range estimates ranged from 2.4 to 35.4 ha on the treated area and 4.6 to 22.9 ha on the untreated area.

4.2. Habitat use

4.2.1. Treated area

During each season habitat use differed from availability ($X^2=307.5$, 277.0, 134.4, 338.4, and

557.9 for breeding season, June, July, August, and summer, respectively, $df=10$, all P 's<0.001). Use of sapling stands of both major species types (aspen-black cherry and mixed oak) was significantly greater than availability in all seasons except August, when they were used proportionally (Table 3). Mixed oak poletimber stands were used more than expected during each month and season (Table 3). Use of mixed oak sawtimber stands increased during each period from the breeding season until August, but were used less than expected based on availability.

During the breeding season, sapling stands accounted for 60% of grouse locations and sawtimber stands were used sparingly. Poletimber-size stands

Table 3

Male ruffed grouse habitat use (proportion of telemetry locations) during the breeding season (capture – May 31) and summer (June 1–August 31) relative to availability on the treated area of State Game Lands 176, Pennsylvania, 1990 and 1991

Habitat type	Proportion of area	Breeding		June		July		August		Summer	
		Use	Pref. ^a	Use	Pref.	Use	Pref.	Use	Pref.	Use	Pref.
Grassy-open	0.04	0.02	0	0.01	–	0.03	0	0.01	–	0.02	–
Aspen-black cherry saplings	0.13	0.30	+	0.28	+	0.25	+	0.08	0	0.23	+
Aspen-black cherry sparse saplings	0.03	0.00	–	0.00	–	0.02	0	0.02	0	0.01	–
Aspen-black cherry poletimber	0.06	0.10	0	0.08	0	0.07	0	0.08	0	0.08	0
Aspen-black cherry sawtimber	0.01	0.00	–	0.01	0	0.00	–	0.00	–	<0.01	–
Scrub oak	0.04	0.02	0	0.04	0	0.06	0	0.01	–	0.04	0
Mixed oak saplings	0.10	0.30	+	0.17	+	0.20	+	0.13	0	0.17	+
Mixed oak sparse saplings	0.07	0.07	0	0.09	0	0.10	0	0.08	0	0.09	0
Mixed oak poletimber	0.02	0.08	+	0.11	+	0.06	+	0.23	+	0.11	+
Mixed oak sawtimber	0.47	0.12	–	0.21	–	0.22	–	0.38	0	0.24	–
Pine-mixed forest	0.03	0.01	–	0.01	–	0.00	–	0.01	0	0.01	–

^a 0, habitat used in proportion to availability ($P>0.0023$); –, habitat used less than availability ($P\leq 0.0023$); +, habitat used more than availability ($P\leq 0.0023$).

Table 4

Male ruffed grouse habitat use (proportion of telemetry locations) during the breeding season (capture – May 31) and summer (June 1–August 31) relative to availability on the untreated area of State Game Lands 176, Pennsylvania, 1990 and 1991

Habitat type	Proportion of area	Breeding		June		July		August		Summer	
		Use	Pref. ^a	Use	Pref.	Use	Pref.	Use	Pref.	Use	Pref.
Grassy-open	0.04	0.01	–	0.02	0	0.01	–	0.02	0	0.02	–
Aspen-black cherry saplings	0.01	0.00	–	0.00	–	0.00	–	0.00	–	0.00	–
Aspen-black cherry sparse saplings	0.01	0.02	0	0.01	0	0.00	–	0.00	–	<0.01	–
Aspen-black cherry poletimber	0.14	0.42	+	0.36	+	0.39	+	0.37	+	0.37	+
Aspen-black cherry sawtimber	0.02	0.10	+	0.10	+	0.10	+	0.02	0	0.10	+
Scrub oak	0.05	0.05	0	0.02	–	0.01	–	0.01	–	0.01	–
Mixed oak saplings	0.01	0.00	–	0.00	–	0.00	–	0.00	–	0.00	–
Mixed oak sparse saplings	0.01	0.00	–	0.00	–	0.00	–	0.00	–	0.00	–
Mixed oak poletimber	0.01	0.02	0	0.04	0	0.01	–	0.01	0	0.02	0
Mixed oak sawtimber	0.62	0.28	–	0.29	–	0.31	–	0.46	–	0.34	–
Pine-mixed forest	0.08	0.10	0	0.11	0	0.17	+	0.11	0	0.13	+

^a 0, habitat used in proportion to availability ($P>0.0023$); –, habitat used less than availability ($P\leq 0.0023$); +, habitat used more than availability ($P\leq 0.0023$).

were used in proportion to availability in the aspen-black cherry type and greater than availability in the mixed oak type in all seasons and sparse sapling stands of both types were usually used in proportion to availability (Table 3).

4.2.2. Untreated area

Habitat use by male ruffed grouse on the untreated area differed from availability in all seasons ($\chi^2=224.4, 246.2, 208.4, 65.5$, and 440.3 , $df=10$, all P 's <0.001 for breeding, June, July, August, and summer, respectively). Use of aspen-black cherry pole- and sawtimber was greater than expected in all seasons except during August when sawtimber was used in proportion to availability (Table 4). Aspen-black cherry pole- and sawtimber stands were the only types used more than expected during the breeding season. Dense sapling stands are usually associated with breeding habitat (Gullion, 1984); however, only 2% of the untreated area was in such stands (excluding scrub oak).

Use of mixed oak sawtimber increased during each period from the breeding season through August, but use was always less than availability (Table 4). Pine and mixed forests were used at least in proportion to availability in all seasons (Table 4). These areas often had dense scrub oak understories that resulted in secure cover for grouse.

5. Discussion

Cumulative summer home range sizes were larger than individual monthly home ranges although the differences were not significant. Male ruffed grouse shifted their home range location through the summer. The shifting of location was likely due to the pattern of availability of summer foods (e.g. *Rubus* sp., *Vaccinium* sp.). The home ranges of male ruffed grouse on SGL 176 were similar to the 6.7 ha in Minnesota during the breeding season (Archibald, 1975). Breeding and summer home ranges on SGL 176 were much smaller than the 68.0 ha (95% harmonic mean) and 45.0 ha (MCP) reported in Missouri (Thompson and Fritzell, 1989), 71.7 ha (MCP, male and female combined) of transplanted grouse in Tennessee (Gudlin and Dimmick, 1984), and 38.1 ha (MCP, adult male) in Tennessee (Epperson, 1988).

Mean home range size tended to be larger on the untreated area than the treated area for each month and season (Table 2), although differences were not significant. Radiocollared grouse did not move between areas.

Male ruffed grouse exhibited distinct trends in use of habitats on SGL 176. Distribution of grouse captures on the untreated area may have biased the results toward selection for aspen-black cherry habitat. However, breeding male censuses indicated that most

drumming males on the untreated area were in aspen-black cherry stands (G.L. Storm unpub. data). Breeding male grouse were more evenly distributed across the treated area than the untreated area. Male grouse used mixed oak stands on the treated area as intensely as aspen-black cherry stands; however we observed little use of mixed oak, relative to aspen-black cherry on the untreated area (Tables 3 and 4). This is likely due to the availability of young, mixed oak stands on the treated area. Only 3% of the untreated area was in sapling and poletimber mixed oak stands, compared to 19% of the treated area. Consequently, densities of breeding males on the treated area have been higher than those on the untreated since 1988, and densities of breeding males in young mixed oak patches have exceeded 25/100 ha (McDonald et al., 1994). Wiggers et al. (1992) also reported that grouse densities in predominantly oak forests in Missouri were positively correlated with the amount of early successional forest in the surrounding landscape.

The understory of aspen-black cherry stands on the untreated area was usually scrub oak or hazel, which provided suitable structure for breeding habitat despite the presence of some large overstory trees. Much of the aspen-black cherry overstory had a thin canopy. The open nature of the overstory allowed the maintenance of a dense understory that compensated for the lack of sapling-sized stands.

Male grouse on both areas tended to use oak sawtimber with increasing intensity as the summer progressed. Mixed oak sawtimber stands had large amounts of blueberry that ripened during August. The availability of ripe fruits may have caused grouse to increase their use of mixed oak sawtimber. Some clearcut areas also contained blueberries but often blackberry or dense saplings precluded growth of blueberries. The greater availability of sapling stands (excluding scrub oak) on the treated area than the untreated area (treated area=33%, untreated area=4%) was accompanied by greater than expected use of these stands.

During all seasons young mixed oak stands on the treated area were used at least in proportion to availability and usually greater than expected based on habitat availability. Mixed oak poletimber stands likely were used more than expected in all periods due to their juxtaposition to breeding cover (mixed oak saplings) during the breeding season and early

summer and sawtimber stands (i.e. food) during August.

The widespread availability of scrub oak as an understory component in aspen-black cherry stands and in pure stands on the untreated area provided habitat structure similar to sapling stands. Scrub oak was not present over large areas of oak sawtimber. With the absence of a dense understory, oak sawtimber had limited usefulness to male ruffed grouse until blueberries began to ripen. However, heavy fall use of acorns by grouse (Servello and Kirkpatrick, 1988) indicates the importance of a sawtimber component in oak forests.

5.1. Management implications

Forest tree species composition does not appear to be a limiting factor to grouse in central Pennsylvania. Managed oak forests sustained higher grouse densities than managed aspen forests (McDonald et al., 1994). Grouse responded to forest structure rather than species composition. Clearcutting mixed oak forests resulted in more forest area suitable for use by male ruffed grouse during the breeding season and summer, thus increasing their density relative to adjacent unmanaged forests. Research in areas of the ruffed grouse's range dominated by aspen forests demonstrated the ability of clearcutting to increase grouse densities and use of those forests (Gullion and Alm, 1983; Kubisiak, 1985). We demonstrate that the same principles can be used successfully in the oak forests typical of the eastern and southern portions of the ruffed grouse's range.

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